

Issue # 7: Numerator Data

Ron Huang

October 17, 2025

1 Data Manipulation

I used three major tables from Numerator original dataset: item-trip level table (item information including quantity, price, product categorization in each trip), item table (categorization and product description), people attributes table (household demographics and other characteristics). Major issues to be noted about the data:

- Demographics:

According to the conversation with Jintaek Song, the demographics may not have been frequently updated across time. For example, income may be variable across time but the record in Numerator may not be up to date. However, we can confirm if this is still the case in the latest data.

- Online receipt:

1) people have to link the email account to Numerator so that Numerator can extract e-receipt from their email account; it is then hard to discern whether, when there is no online transaction, it is because people did not buy anything online or it is because of missing receipts where people did not link the email account to Numerator.

2) SNAP indicator could be inaccurate.

However, these setup and problem may have changed over time and we need to confirm with Numerator.

- SNAP Indicator:

The SNAP indicator is at the receipt level, which means that we cannot distinguish SNAP-eligible foods paid by EBT and cash.

- Compensation Scheme:

Numerator's monetary reward to receipts varies with receipt type. Grocery receipts always receives reward (reward value increasing with transaction value), whereas others are entered into lottery so only a portion will receive reward. Still, we need to confirm with Numerator if this is the setting nowadays.

- No serving size information and missing UPC/GTIN codes

Around half of items in Numerator do not have UPC/GTIN code that can be linked to USDA or other serving/package size dataset. Two workarounds:

1) Song’s paper suggests creating crosswalks between Nielsen, which can be linked with USDA by UPC, and Numerator by matching or creating the common food categories. We can use Nielsen data to convert product-level information to the crosswalk level by computing an expenditure-weighted average of serving/package size per dollar for each new matched category (crosswalk) so that we can use this compressed information to compute serving size by multiply item unit price in Numerator by the serving/package per dollar linked through this crosswalk. This method assumes same serving/package size per dollar distribution in the shared food category between Nielsen and Numerator.

2) impute missing UPC/GTIN by item unit price with similar products that have UPC in Numerator data.

3) extract serving/package size from ‘**product description**’ column in Numerator data. A quick inspection into the ‘**product description**’ column suggests that food without UPC/GTIN codes in Numerator may usually have the serving size and package size included in the ‘**product description**’ column. However, it is now hard to get an exact sense if this applies for all food items we are interested in and how often we can do this extraction, because the dataset is large, how serving/package size is labeled could be different across products, and the ‘**product description**’ column is string so it will take time to parse.

I created monthly household expenditure table based on the following steps:

1. selected only static panelists for a period of two years (21-22 and 23-24) and merged item-trip, item, and people attribute tables by item ID and household ID (all left merge with item-trip table).
2. created food at home and food away from home indicator (based on categorization in Table 16)
3. created soda indicator (based on categorization in Table 17)
4. created SNAP eligible indicator (alcoholic beverage and Deli prepared food as ineligible items; this is a rough classification based on the ‘**department**’ label in the data; we can use ‘**category**’ label to make a finer classification manually)
5. created **SNAP participation indicator** based on:
 - 1) whether SNAP benefit is ever used during the month
 - 2) $\frac{\text{EBT spending}}{\text{Food Spending}} > 0.1$ in the month
 - 3) household income below threshold in Figure 15, though not sure if the reported income is for household or primary shopper; we choose \$100k as a conservative upper bound to be on SNAP.

6. **SNAP treatment indicator:** giving the above conditions for SNAP participation indicator in one month, we define a household to be treated during the period of our analysis if the participation rate is larger than 0.5, which means that a household needs to be on SNAP, in our definition, for more than 12 months out of 24 months. We can adjust this threshold for to be treated to be smaller like ‘at least 2 or 3 months satisfying these conditions’ but there is a tradeoff that the treated group is going to include more households with lower participation rate in general, which can confound the SNAP group. We can also make this not mutually exhaustive; for example, the SNAP participation rate below 0.2 is control and above 0.7 is treated to exclude those floating in the middle who we are more unsure about, but this will also decreases the sample size

7. Monthly expenditure variable defined (EBT related spendings are conditional on SNAP household):

$$\text{food/soda} \left\{ \begin{array}{l} \text{food/soda spending} \\ \text{food/soda spending by EBT} \\ \text{food/soda at home spending} \\ \text{food/soda at home spending by EBT} \\ \text{food/soda away from home spending} \\ \text{food/soda away from home spending by EBT} \quad (\text{for sanity check}) \end{array} \right.$$

8. major demographics: income bucket, age bucket, education bucket, household size, number of children bucket, ethnicity, employment, marital status, zip code/state

- for binned variables, I took the mean of the upper and lower bounds.

2 Quality Check

2.1 SNAP Indicator and EBT Spending

Figure 1 shows the correlation between monthly EBT spending conditional on households with ≥ 12 consecutive months of SNAP participation in 2023-2024. Typically people would spend all of the SNAP benefit, and therefore the correlation between monthly EBT spending should be close to 1 with high R-squared. The coefficient here is around 0.65 with R-squared 0.44 suggesting that there could be under-reporting of receipts with SNAP benefit used (with the under-reporting varying over time within households). As discussed previously, this weaker correlation could also be due to:

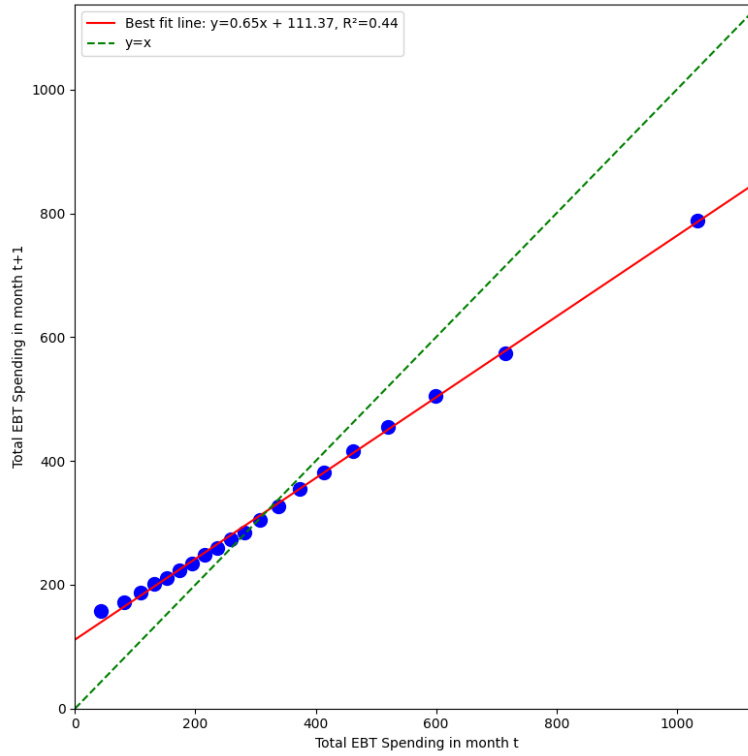


Figure 1: EBT Spending Correlation between Month t and $t+1$ in 2023-2024 (conditional on household with ≥ 12 consecutive months of SNAP participation indicator=1)

- overestimation of EBT spending as we cannot distinguish between food at home paid by cash and those paid by EBT on one receipt
- the fact that we are using calendar month but SNAP benefit in certain states can be distributed across the month (we can limit to those distributed at the beginning of the month $\rightarrow$ work in progress)

Note that in the previous version the coefficient was around 0.8, which was an artifact of a coding error.

Figure 2 shows the proportion of household monthly food spending paid by EBT card. This is a potential way of identifying inframarginal SNAP household. The bunching of households at proportion of 1 represent households replying entirely on SNAP benefit for food, while inframarginal households are those who can afford more on food aside from using SNAP benefit (though proportion below 1 can be confounded by missing receipts). The mean of this distribution of proportion of food spending paid by EBT is about 50% and median is 48%, which is about 8-10% higher than the 40% reported by Hasting-Shapiro 2018 paper. Recall that here we restricts to households with ≥ 12 consecutive SNAP spell, where in each month we define a household is on SNAP if SNAP benefit is ever used and the ratio of EBT spending to food at home spending is larger than 0.1. In Table 10, this proportion is about 44% for all who are SNAP treated in our definition (SNAP participation rate > 0.5). Therefore, it is worth noting that

this proportion is higher in our case than the estimate in Hasting-Shapiro paper even though we are under the risk of under-reporting of SNAP benefit. However, this could be also due to the fact that we may overestimate EBT spending by counting food at home paid by cash and EBT card both as EBT spending.

Figure 3 shows the distribution of consecutive SNAP spell for all households on SNAP. This intends to replicate the 6-month divisibility in Figure 2 of the Hasting-Shapiro paper. We should be able to observe the same bunching effect at every six month if the data captures all EBT carding spending.

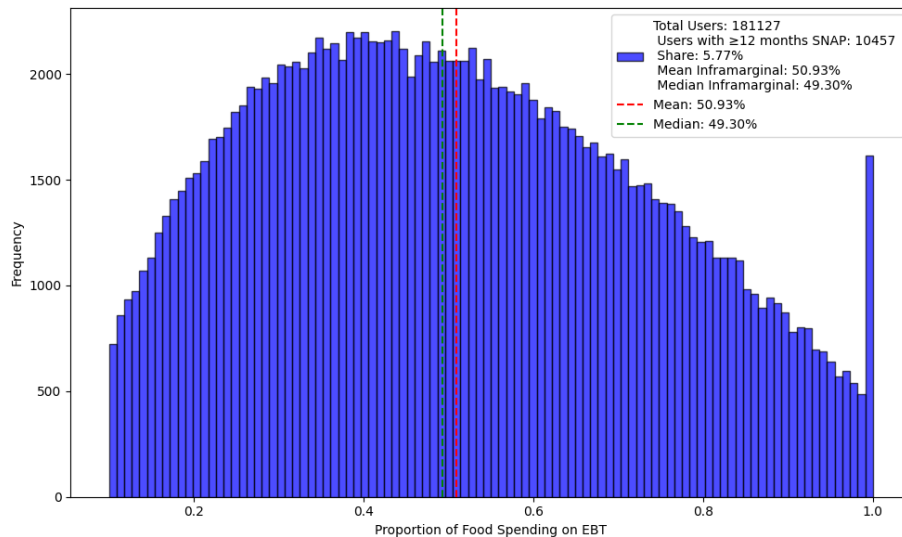


Figure 2: Proportion of Food Spending by EBT conditonal (conditional on household with ≥ 12 consecutive months of SNAP participation indicator=1)

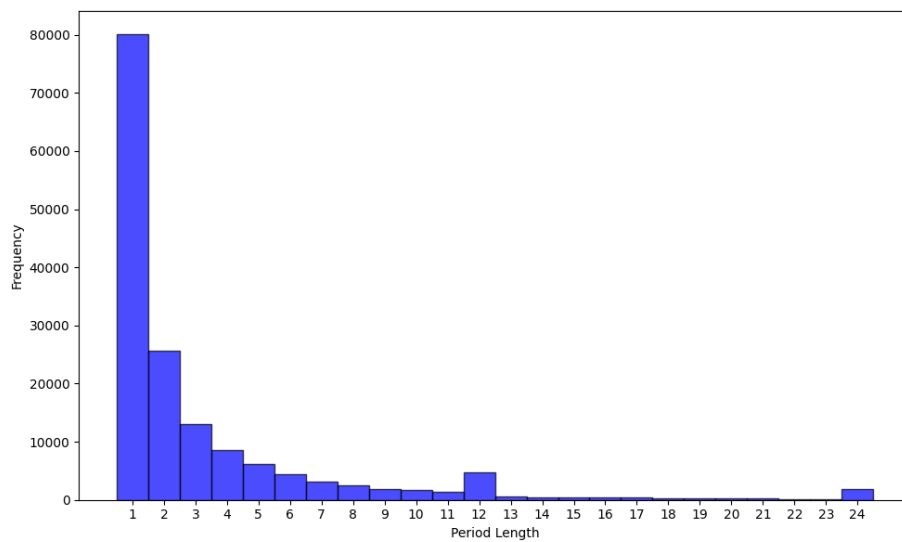


Figure 3: Distribution of Consecutive SNAP Spells

Table 1: Annual Expenditure in 2023

	Numerator				CEX			
	< 20,000	20,000 to 29,999	30,000 to 39,999	Total	<15,000	15,000 to 29,999	30,000 to 39,999	Total
Total Spending	14,921.84 (21,658.75)	14,130.09 (19,978.39)	13,907.81 (18,461.29)	14,414.83 (20,325.43)	32,081 (1,385.24)	35,587 (1054.11)	47,027 (1212.88)	38,231
Food Spending	6,297.14 (9,181.03)	6,007.39 (9,336.73)	5,764.50 (7,939.10)	6,064.11 (8,879.72)	5,684 (328.22)	5,181 (233.61)	6,665 (323.48)	5,843
Food at home	5,674.39 (8,486.89)	5,428.77 (8,824.30)	5,193.74 (7,346.39)	5,468.28 (8,264.27)	4,037 (199.09)	3,624 (185.00)	4,430 (286.97)	4,030
Food away from home	622.76 (1,244.22)	578.63 (1,173.66)	570.76 (1,084.51)	595.83 (1,180.79)	1,647 (189.40)	1,556 (96.23)	2,235 (135.78)	1812
Soda spending/nonalcoholic beverage	263.68 (587.44)	258.81 (512.47)	239.28 (448.09)	255.22 (530.10)	490 (36.54)	434 (33.79)	514 (38.74)	479
Prop. of total spending on food	44% (0.14)	44% (0.13)	43% (0.13)	44% (0.13)	17.7%	14.6%	14.2%	15.5%
Prop. of food on FAH	90% (0.10)	90% (0.10)	90% (0.10)	90% (0.10)	71%	69%	66%	68%
Prop. of food on FAFH	10% (0.10)	10% (0.10)	10% (0.10)	10% (0.10)	29%	30%	34%	31%
Prop. of Soda/nonalcoholic beverage	4% (0.04)	5% (0.05)	4% (0.04)	4% (0.04)	8%	7%	8%	7.6%

Notes: This table shows the average annual spending and proportions in 2023. The soda spending row records soda spending in Numerator data, and nonalcoholic beverage spending in CEX data. Note that the proportion and total column for CEX data are computed as the proportion and average of the corresponding aggregated spending variables above.

2.2 Comparison to CEX Expenditure by Income

Table 1 compares the average annual expenditure of household in 2023 between Numerator and CEX data. Major observations if taking CEX as the ground truth:

- $\downarrow$ Total spending: Numerator data significantly underestimates the total spending. However, this makes sense as Numerator emphasizes the most on grocery, or food and food at home specifically, receipts, which is reflected by the Numerator's compensation scheme for rewarding receipt upload, where grocery receipts always receive monetary reward, whereas other receipts are entered into lottery.
- $\approx$ Food spending/Food at home: The difference in annual averages of household food spending and food at home spending is much smaller than that of the total spending, for the similar reason as above.
- $\downarrow$ Food away from home: Similarly, Numerator captures less food away from home than food at home spending for the same reason as above. Besides, Numerator does not capture full restaurant services well, while it may capture some of the limited restaurant services and online food delivery that are potentially food away from home.
- $\downarrow \frac{FAFH}{FAH}$: the ratio of food away from home to food at home in Numerator is around 1 to 9 (0.1), whereas the ratio in CEX is about 3 to 7 (0.43) (for low income group).
- $\uparrow$ Variation: The standard deviations of all dollar estimates are much larger for Numerator than for CEX.

Aside from those major observations for low income group, for high income group (Table 21), the estimates of all types of annual averages would be worse (more significantly underestimated) as there could be a lack of incentives for high-income group to upload receipts to Numerator, but they may not be the major population of interested as they would not be eligible for SNAP.

2.3 Other Balance Check

The static panel in the Numerator Datasets is structured as follows:

- Each cohort in the panel represents a set of households with a 12-month spells of high-quality purchases record and matches the national demographic composition.
- Hence, in a period of one calendar year, there is only one cohort, which has the spell starts on Jan of the year, covers the entire calendar year.
- One household could be in multiple cohorts, which means that it could be consistently included in the static panel for more than 12 months. For example, in a

period of 2-calender year, one household could stay in the static panel by entering 12 different cohorts (from cohort starts in Jan to cohort starts in Dec of the first calender year).

- There are three weights in the data, two of which are related to the static panel, the demographic weights and national projection factor. Both of them are assigned to each household within each cohort and are invariance across the period covered by the cohort. The demographic weight is to balance the sample to match the demographics at the national level, and the national projection factor is to project the levels of variables to the national level. The sum of demographics weights equals to the number of households in each cohort.
- If we want to, for example, conduct analysis for 2 calender years, we need to restrict the sample to two cohorts, one starting from Jan of first year and one starting form Jan of second year, in order to correctly apply the weights. However, this will create an unbalanced panel, as Numerator replaces about 30-40% of households between two adjacent cohorts (adjacent cohorts as the two that have start date 12 months apart from each other)
- The sample size in the static sample is 200,000 households stably across cohorts. Note that the cross-section sample size of the panelists in each month may not be stable around 200,000

Figure 4 verifies the sample size of the static panelists and rate of changes of panelists between cohorts and months. In Panel A, around 30-40% of households are replaced with new ones; in panel B, the net attrition rate between monthly sample (not between cohort) is around 3-4%; in panel C, the cohort size is stable at 200,000 starting 2023; Panel D verifies that the number of dropouts equals to the number of addition between two cohort starting 2020.

3 Summary Statistics

3.1 Summary Tables

Table 4 shows that breakdown of number of households by states. Note that the total number does not sums up to 200,000 because we limit to 50 states + DC, while there are some panelists live in some U.S. territories or other zip-code regions that in 50 state + DC

Figure 4: Net Changes

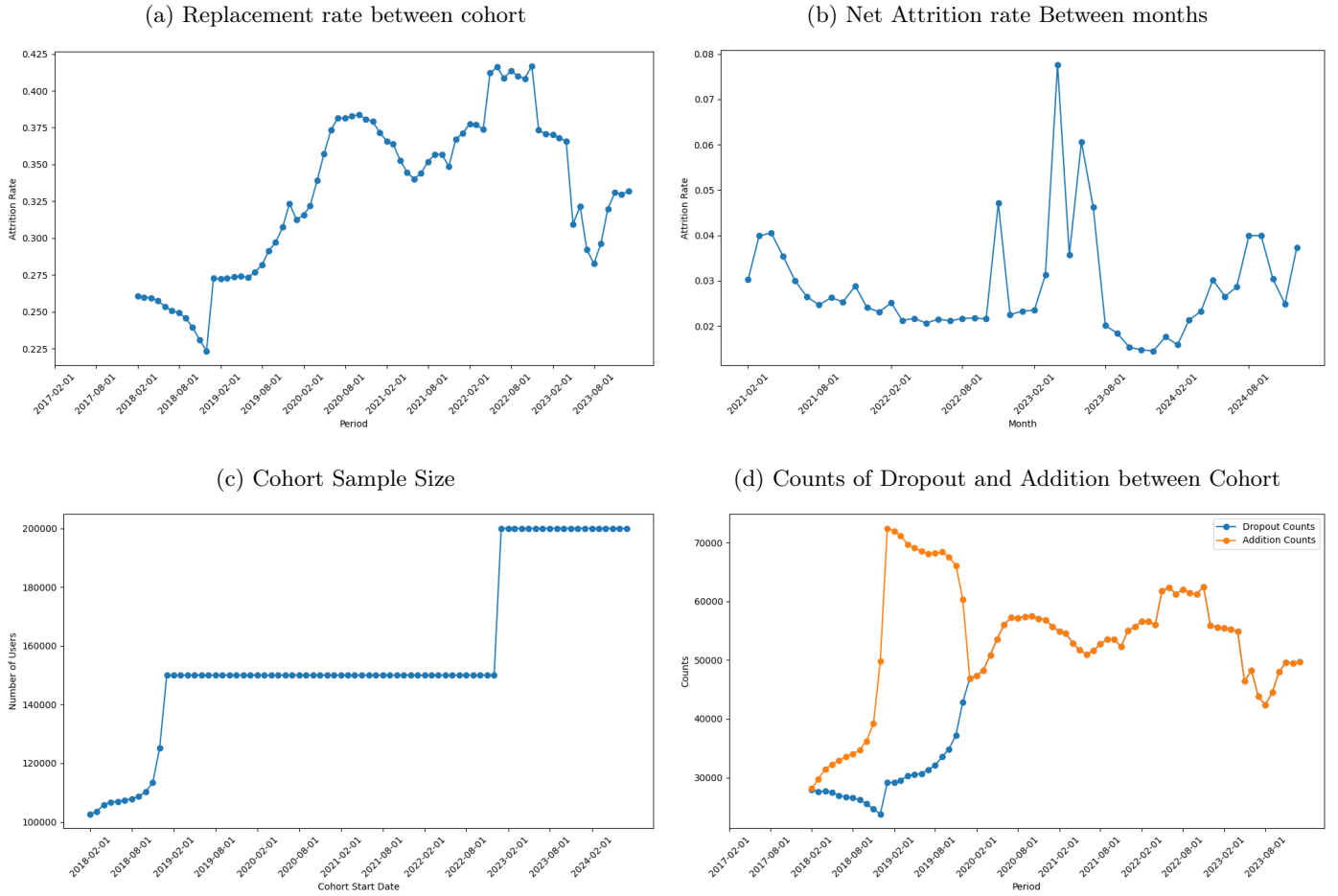


Table 3: Number of Users in Affected and Unaffected States

State	2023	2024
Household affected in Jan 2026		
FL	15719	16075
IA	2084	2142
ID	1046	1006
IN	4981	4852
LA	2814	2772
NE	1333	1360
OK	2211	2190
UT	1533	1523
WV	1421	1300
Subtotal	33142	33220
Household affected after Jan 2026		
AK	378	392
CO	2850	2880
TX	16649	16703
Subtotal	19877	19975
Treated States	53019	53195
Control States	146557	146550
Total	199576	199745

Notes: In static panel, each static period is defined by a 12-month window, and the panel is balanced within each static period but may not be balanced between static period. Here is total number does not sums up to 200,000 because we limit to 50 states + DC, while there are some panelists live in some U.S. territories or other zip-code regions that in 50 state + DC

Table 4: User-Month Panel - Food Spending Percentiles

percentile	food spending 2023	food spending 2024
50	315.24	377.27
60	409.35	471.75
70	548.28	604.43
80	779.67	823.70
90	1265.30	1268.08
95	1883.47	1795.01
96	2112.78	1985.96
97	2433.36	2253.95
98	2926.79	2675.30
99	3862.00	3484.38
99.5	4875.17	4400.69
100	97048.25	57237.37

Table 5: User-Month Panel - Soda Spending Percentiles

percentile	soda spending 2023	soda spending 2024
50	6.61	8.27
60	10.66	13.05
70	16.58	19.96
80	26.62	31.15
90	49.41	54.92
95	81.87	86.16
96	94.74	97.91
97	113.70	114.66
98	144.35	141.13
99	208.35	194.93
99.5	287.51	260.49
100	28746.10	8997.98

Table 4 and 5 shows values of food and soda expenditure at each percentile. We can observe that there are outliers, which typically exist beyond the 99.5 percentile. For the trend and regression analyses in the later section, we will winsorize these variable at the 99.5 percentile. For the summary statistics, we do not winsorize them for now.

Roadmap: The following set of tables show the summary statistics of spending variables.

Table 6 shows the demographics by treated and untreated states; average household size of all sample is about 2.8, average age is 50.22, 74% of household (primary shopper) is White, 9% is Black, 2% is Asian, and 13% is Hispanic.

Table 7 shows the expenditure of food by treated and untreated states; average total spending is 1461.19 dollars, average food spending is 575.88 dollars, about 44% of total spending are on food, 90% of food are on food away from home, 40% of food are paid by EBT among SNAP households, and monthly SNAP participation rate is about 12% (previously around 17% without conditioning on ratio of EBT spending to food at home

spending larger than 10%).

Table 8 shows the soda spending by treated and untreated state; households spend on average 21 dollars on soda, with 4% of food spending are on soda, 87% of soda are at home consumption, 5% of EBT spending are on soda among SNAP households, and 37% of soda are paid by EBT among SNAP households.

Table 9, 10, and 11 show the same set of statistics by SNAP and Non-SNAP households instead. SNAP households earn around 33798 dollars whereas non-SNAP households earn around 93424 dollars on average annually. 27% versus 60% of primary shoppers in SNAP and non-SNAP households are married. 31% versus 24% of SNAP and non-SNAP households have children. Monthly SNAP participation rate is 79% for SNAP household and 5\$ for non-SNAP households (recall that our SNAP treatment status is not only based on whether SNAP benefit was ever used during the month). SNAP households spend 23.56 dollars on average for soda, whereas non-SNAP households spend 20.77 on average.

In Appendix, Table 18, 19, and 20 show the same set of tables for diabetes status.

Table 6: User-Month Panel - Demographics

		Mean (SD)		Min	Max
	Treated States	Control States	Total		
Income	82536.03 (59720.37)	88327.63 (63969.22)	87365.14 (63319.59)	10000.00	275000.00
Household Size	2.73 (1.43)	2.81 (1.46)	2.79 (1.45)	1.00	7.00
Age	51.60 (13.10)	49.94 (12.92)	50.22 (12.97)	19.00	67.50
Married	0.58 (0.49)	0.56 (0.50)	0.57 (0.50)	0.00	1.00
Education (years)	13.82 (1.34)	13.87 (1.37)	13.86 (1.37)	10.00	16.00
White	0.74 (0.44)	0.69 (0.46)	0.70 (0.46)	0.00	1.00
Black	0.09 (0.28)	0.11 (0.31)	0.10 (0.31)	0.00	1.00
Asian	0.02 (0.15)	0.07 (0.26)	0.06 (0.24)	0.00	1.00
Hispanic	0.13 (0.33)	0.12 (0.32)	0.12 (0.32)	0.00	1.00
Ever Child	0.23 (0.42)	0.25 (0.43)	0.25 (0.43)	0.00	1.00
Number of States	9	42	51		
Number of Users	44,659	221,747	266,406		
Average Number of Months per User	17.83	18.02	17.99		

Notes: Income's 'less than 20000' category is coded as 10000, and 'more than 250000' is coded as 275000. Other income buckets are coded as the mean of a bucket. Education's 'advanced' category is coded as 16, and 'less than high school' is coded as 10.

Table 7: User-Month Panel - Spending

	Treated States	Mean (SD) Control States	Total	Min	Max
Food Spending	548.67 (705.61)	581.30 (794.36)	575.88 (780.40)	0.00	97048.25
Total Spending	1408.17 (2150.18)	1471.76 (2083.31)	1461.19 (2094.71)	0.00	313100.95
Prop. of Total Spending on Food	0.44 (0.18)	0.44 (0.18)	0.44 (0.18)	0.00	1.00
Prop. of Food Spending on FAH	0.90 (0.13)	0.90 (0.13)	0.90 (0.13)	0.00	1.00
Prop. of Food Spending on FAFH	0.10 (0.13)	0.10 (0.13)	0.10 (0.13)	0.00	1.00
Prop of Food on EBT	0.40 (0.23)	0.40 (0.23)	0.40 (0.23)	0.10	1.00
Prop. of EBT Spending on FAH	1.00 (0.01)	1.00 (0.02)	1.00 (0.02)	0.00	1.00
Prop. of EBT on FAFH	0.00 (0.01)	0.00 (0.02)	0.00 (0.02)	0.00	1.00
Ever SNAP	0.11 (0.32)	0.12 (0.33)	0.12 (0.33)	0.00	1.00
Number of States	9	42	51		
Number of Users	44,659	221,747	266,406		

Notes: Prop. of food spending is calculated out of total spending. Prop. of FAH/FAFH spending and food on SNAP is calculated out of total food spending. Prop. of FAH/FAFH on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization. EBT-related outcomes are conditional on households on SNAP.

Table 8: User-Month Panel - Soda Spending

	Treated States	Mean (SD) Control States	Total	Min	Max
Soda Spending	21.22 (44.16)	21.02 (55.45)	21.05 (53.74)	0.00	28746.10
Prop. of Food Spending on Soda	0.04 (0.06)	0.04 (0.05)	0.04 (0.06)	0.00	1.00
Prop. of Soda Spending on Soda at Home	0.88 (0.26)	0.87 (0.27)	0.87 (0.27)	0.00	1.00
Prop. of Soda Spending on Soda Away From Home	0.12 (0.26)	0.13 (0.27)	0.13 (0.27)	0.00	1.00
Prop of EBT Spending on Soda	0.05 (0.09)	0.05 (0.09)	0.05 (0.09)	0.00	1.00
Prop of Soda Spending on EBT	0.37 (0.36)	0.37 (0.37)	0.37 (0.37)	0.00	1.00
Number of Users	44,659	221,747	266,406		

Notes: Prop. of soda spending is calculated out of total food spending. Prop. of soda FAH/FAFH spending is calculated out of total food spending. Prop. of soda on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 9: User-Month Panel - Demographics (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Income	33798.94 (24683.46)	93424.93 (63503.49)	87365.14 (63319.59)	10000.00	275000.00
Household Size	2.92 (1.68)	2.78 (1.42)	2.79 (1.45)	1.00	7.00
Age	48.74 (13.27)	50.38 (12.92)	50.22 (12.97)	19.00	67.50
Married	0.27 (0.44)	0.60 (0.49)	0.57 (0.50)	0.00	1.00
Education (years)	13.22 (1.40)	13.93 (1.34)	13.86 (1.37)	10.00	16.00
White	0.51 (0.50)	0.72 (0.45)	0.70 (0.46)	0.00	1.00
Black	0.21 (0.41)	0.09 (0.29)	0.10 (0.31)	0.00	1.00
Asian	0.04 (0.20)	0.07 (0.25)	0.06 (0.24)	0.00	1.00
Hispanic	0.22 (0.41)	0.11 (0.31)	0.12 (0.32)	0.00	1.00
Ever Child	0.31 (0.46)	0.24 (0.43)	0.25 (0.43)	0.00	1.00
Number of States	51	51	51		
Number of Users	28,513	237,893	266,406		
Average Number of Months per User	27.93	16.80	17.99		

Notes: Income's 'less than 20000' category is coded as 10000, and 'more than 250000' is coded as 275000. Other income buckets are coded as the mean of a bucket. Education's 'advanced' category is coded as 16, and 'less than high school' is coded as 10.

Table 10: User-Month Panel - Spending (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Food Spending	527.37 (760.61)	581.37 (782.42)	575.88 (780.40)	0.00	97048.25
Total Spending	1176.74 (1729.51)	1493.37 (2129.69)	1461.19 (2094.71)	0.00	313100.95
Prop. of Total Spending on Food	0.49 (0.18)	0.43 (0.18)	0.44 (0.18)	0.00	1.00
Prop. of Food Spending on FAH	0.90 (0.12)	0.90 (0.13)	0.90 (0.13)	0.00	1.00
Prop. of Food Spending on FAFH	0.10 (0.12)	0.10 (0.13)	0.10 (0.13)	0.00	1.00
Prop of Food on EBT	0.44 (0.23)	0.31 (0.20)	0.40 (0.23)	0.10	1.00
Prop. of EBT Spending on FAH	1.00 (0.02)	1.00 (0.02)	1.00 (0.02)	0.00	1.00
Prop. of EBT on FAFH	0.00 (0.02)	0.00 (0.02)	0.00 (0.02)	0.00	1.00
Ever SNAP	0.79 (0.41)	0.05 (0.21)	0.12 (0.33)	0.00	1.00
Number of States	51	51	51		
Number of Users	28,513	237,893	266,406		

Notes: Prop. of food spending is calculated out of total spending. Prop. of FAH/FAFH spending and food on SNAP is calculated out of total food spending. Prop. of FAH/FAFH on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 11: User-Month Panel - Soda Spending (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Soda Spending	23.56 (51.58)	20.77 (53.97)	21.05 (53.74)	0.00	28746.10
Prop. of Food Spending on Soda	0.05 (0.06)	0.04 (0.05)	0.04 (0.06)	0.00	1.00
Prop. of Soda Spending on Soda at Home	0.90 (0.23)	0.87 (0.27)	0.87 (0.27)	0.00	1.00
Prop. of Soda Spending on Soda Away From Home	0.10 (0.23)	0.13 (0.27)	0.13 (0.27)	0.00	1.00
Prop of EBT Spending on Soda	0.05 (0.09)	0.04 (0.09)	0.05 (0.09)	0.00	1.00
Prop of Soda Spending on EBT	0.42 (0.37)	0.28 (0.34)	0.37 (0.37)	0.00	1.00
Number of Users	28,513	237,893	266,406		

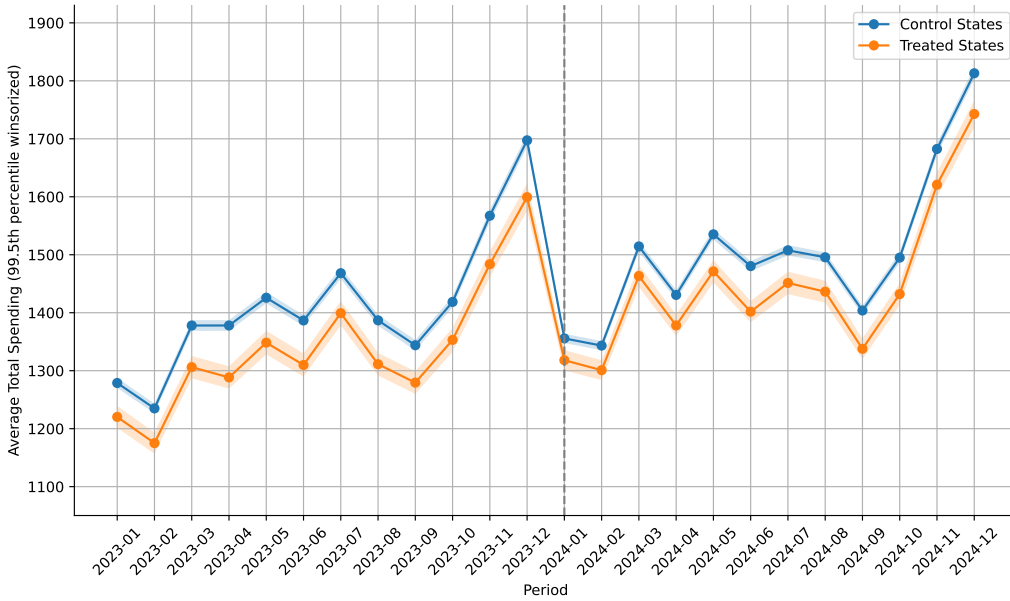
Notes: Prop. of soda spending is calculated out of total food spending. Prop. of soda FAH/FAFH spending is calculated out of total food spending. Prop. of soda on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

4 Trend Analysis

Roadmap: Figure 5 shows the trend of total spending and food spending; there are strong seasonal effects around December of the years, where the total spending surges; the food spending increases less significantly, as we can see in the proportion of total spending on food in the panel b of Figure 7, where the proportion of total spending on food drops during Decembers, though the levels of both are increasing during that period of time. Figure 6 breaks down food spending into food away and food at home spending, where we can see that the seasonal effect is more conspicuous for food at home than food away from home. Panel a and b of Figure 7 shows the trend of food spending by EBT; the decreasing trend at the beginning of the period reflect the reduction in SNAP benefit starting Jan 2023 we discussed previously. Panel c shows the proportion of total spending on food. Panel d shows the proportion of food spending by EBT. Figure 8 shows the trend of soda spending and soda spending paid by EBT card. Figure 9 shows the proportion of food spending on soda, proportion of EBT cash spent on soda, and proportion of soda spending paid by EBT. Figure 10 shows the food/soda spending by SNAP and state treatment status (trends for DDD).

Figure 5: Monthly Average of Total and Food Spending

(a) Total Spending - level (dollars)



(b) Food Spending - level (dollars)

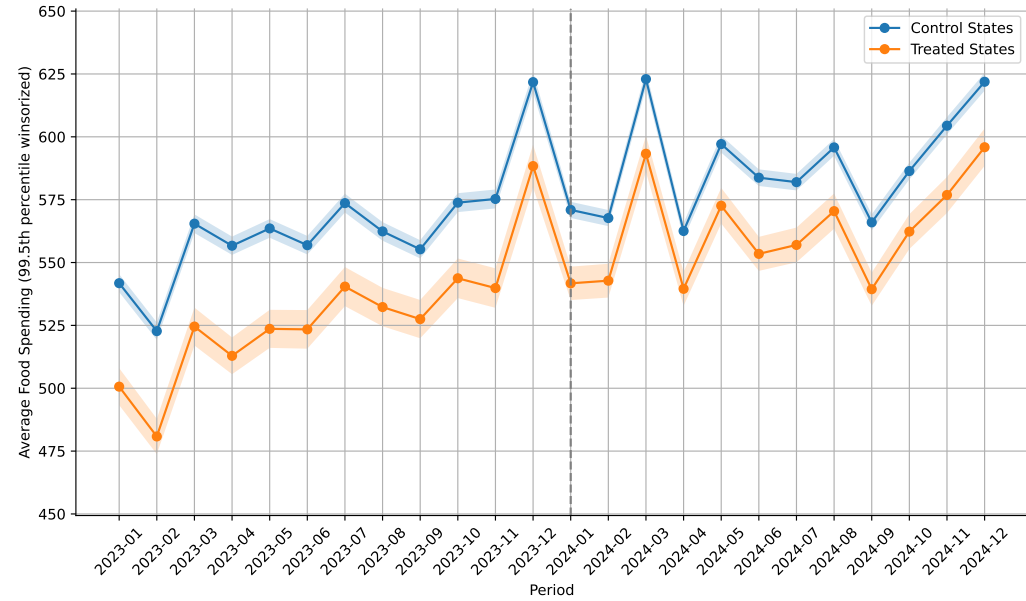
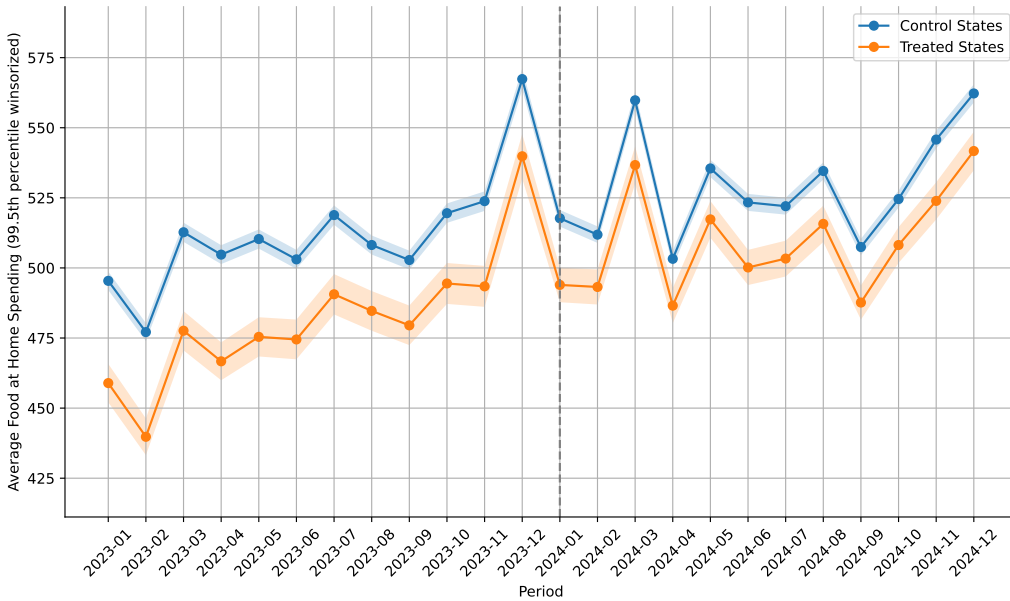


Figure 6: Monthly Average of Food at home and Food away from home

(a) Food at home - level (dollars)



(b) Food away from home - level (dollars)

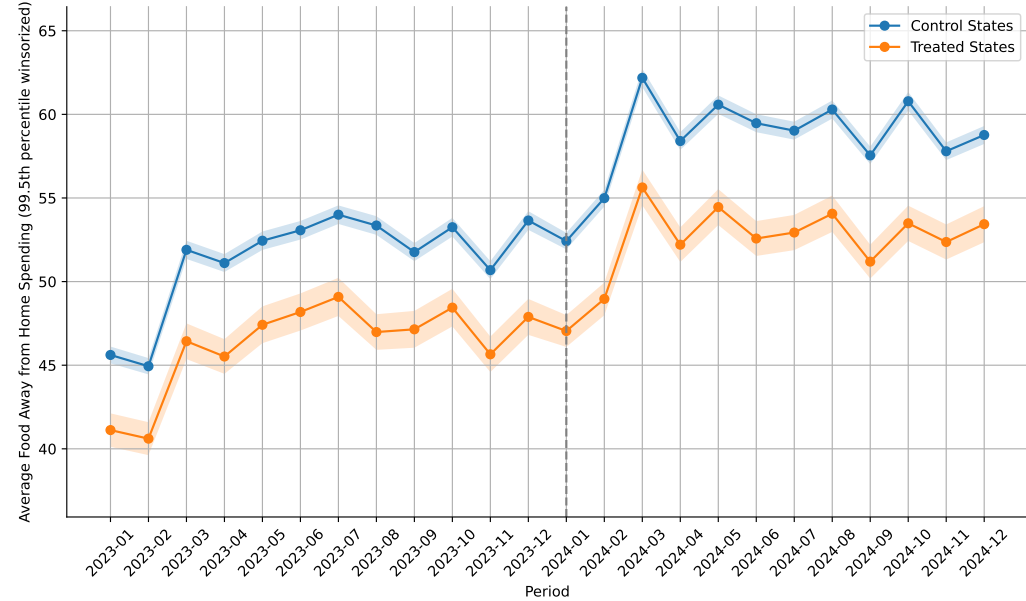
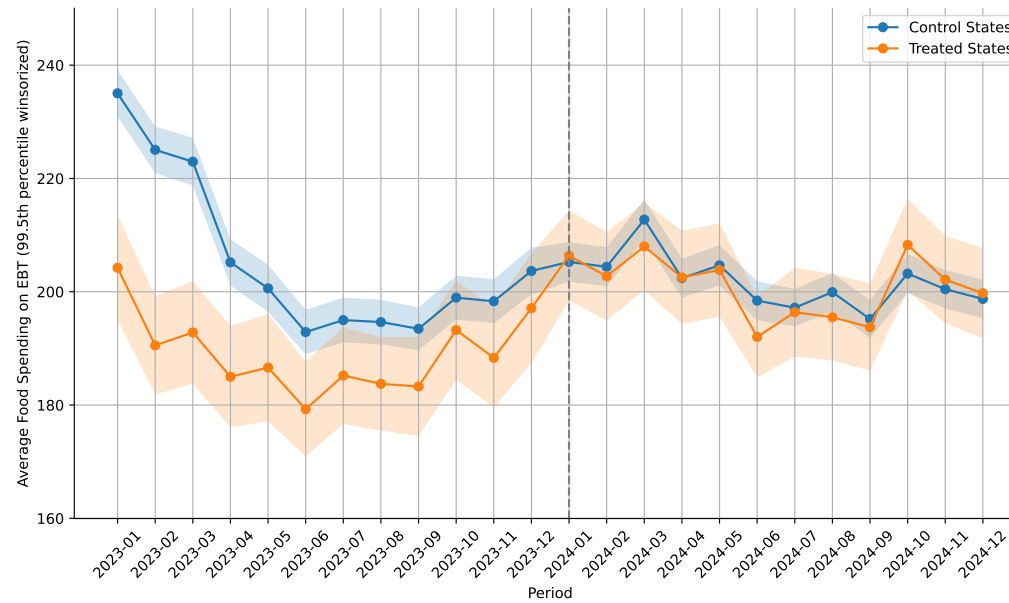
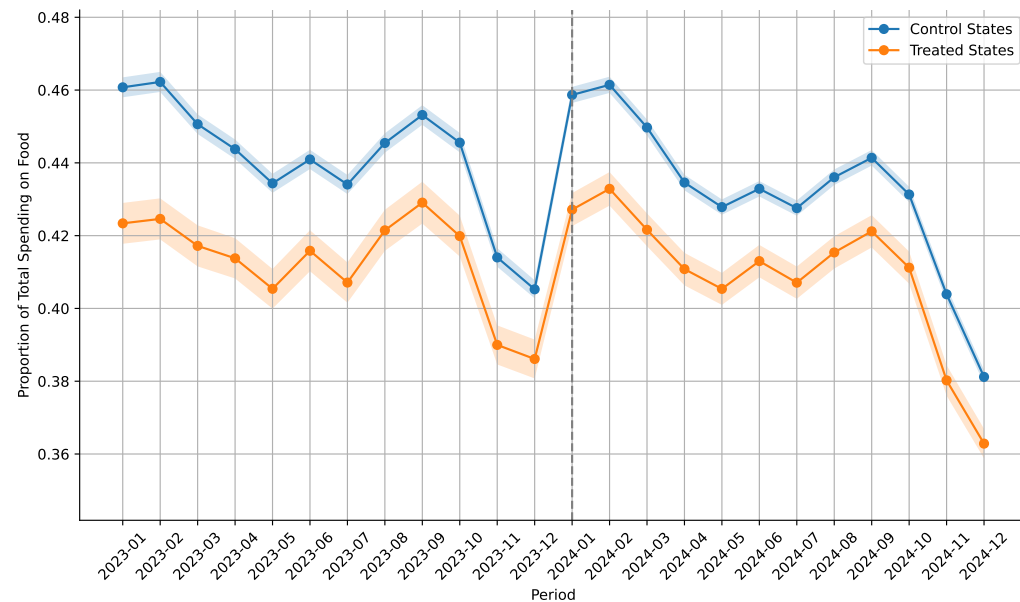


Figure 7: Monthly Average of Food Spending by EBT and Proportion of Food by EBT

(a) Food paid by EBT - level (dollars)



(b) Proportion of Total Spending on Food



(c) Proportion of Food Spending by EBT

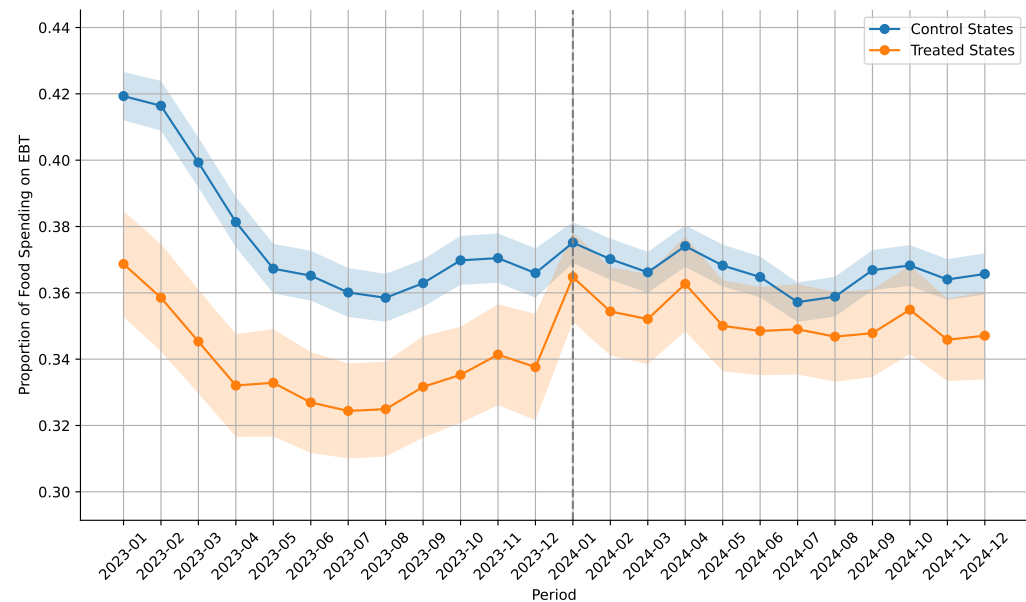
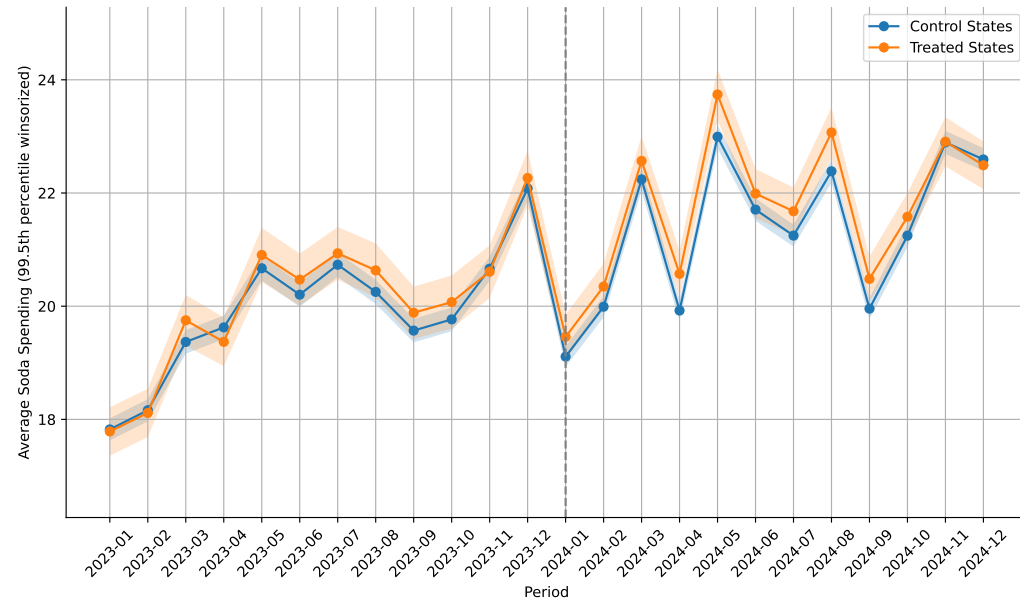


Figure 8: Monthly Average of soda spending and soda spending on EBT

(a) Soda - level



(b) Soda by EBT - level

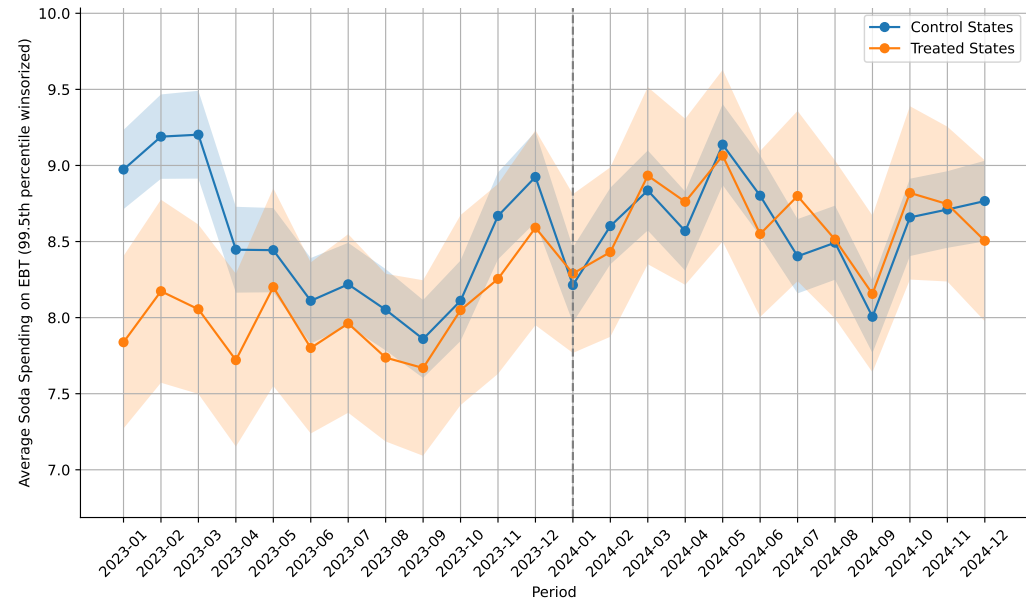
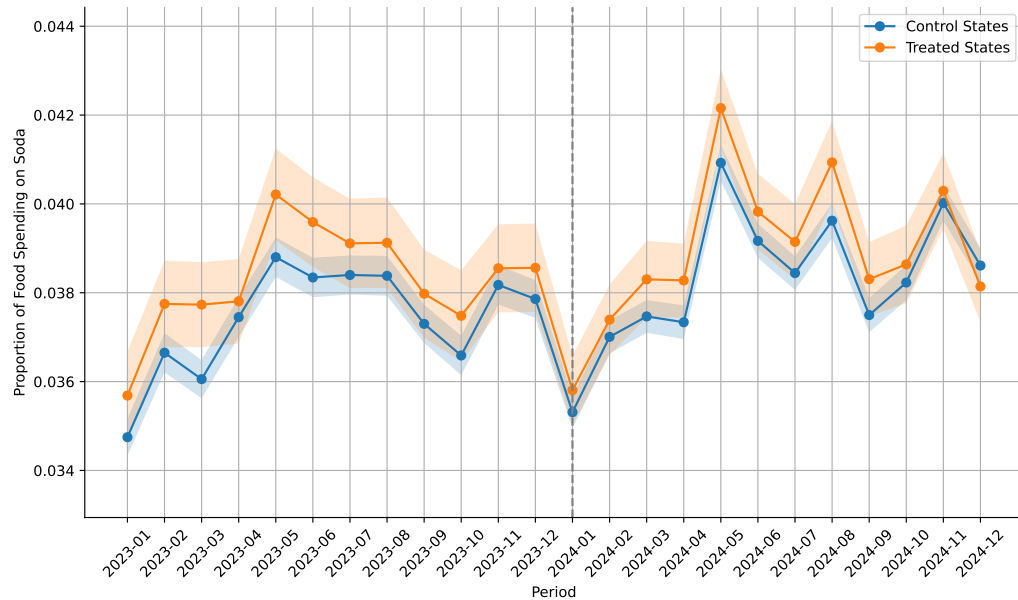
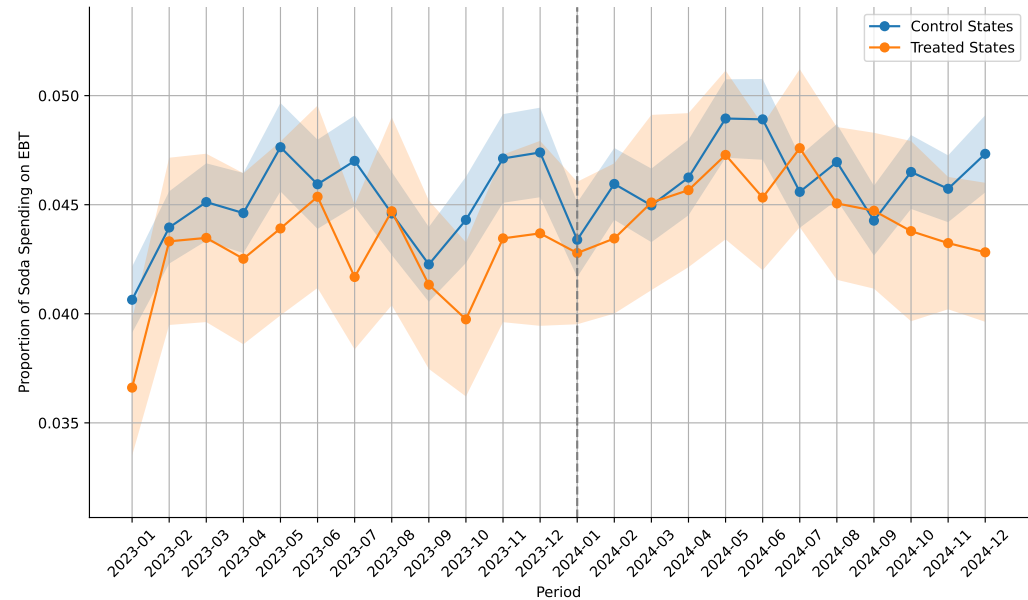


Figure 9: Proportion of Soda Spending

(a) Proportion of Total Food Spending on Soda



(b) Proportion of EBT Spending on Soda



(c) Proportion of Soda Spending by EBT

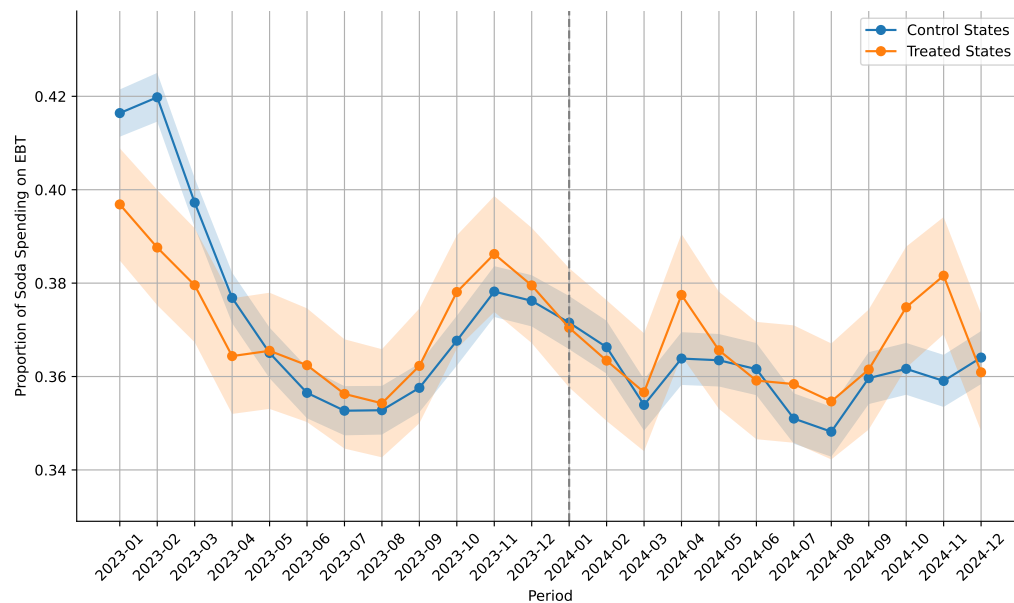
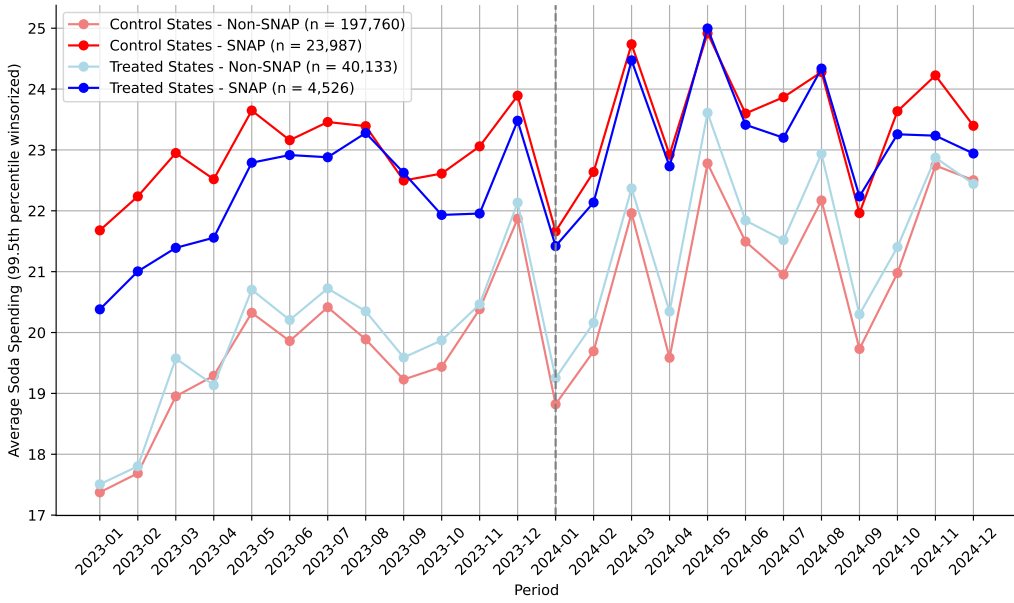


Figure 10: Soda Spending - DDD trends

(a) Soda Spending - level (dollars)



(b) Food Spending - level (dollars)

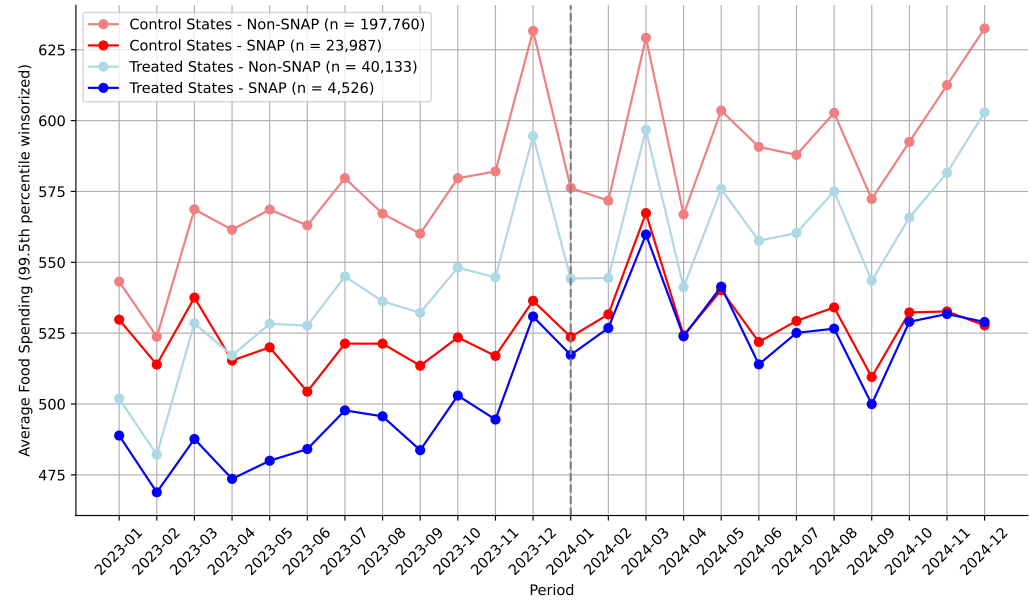


Figure 11: Event Study - TWFE (State Treatment)

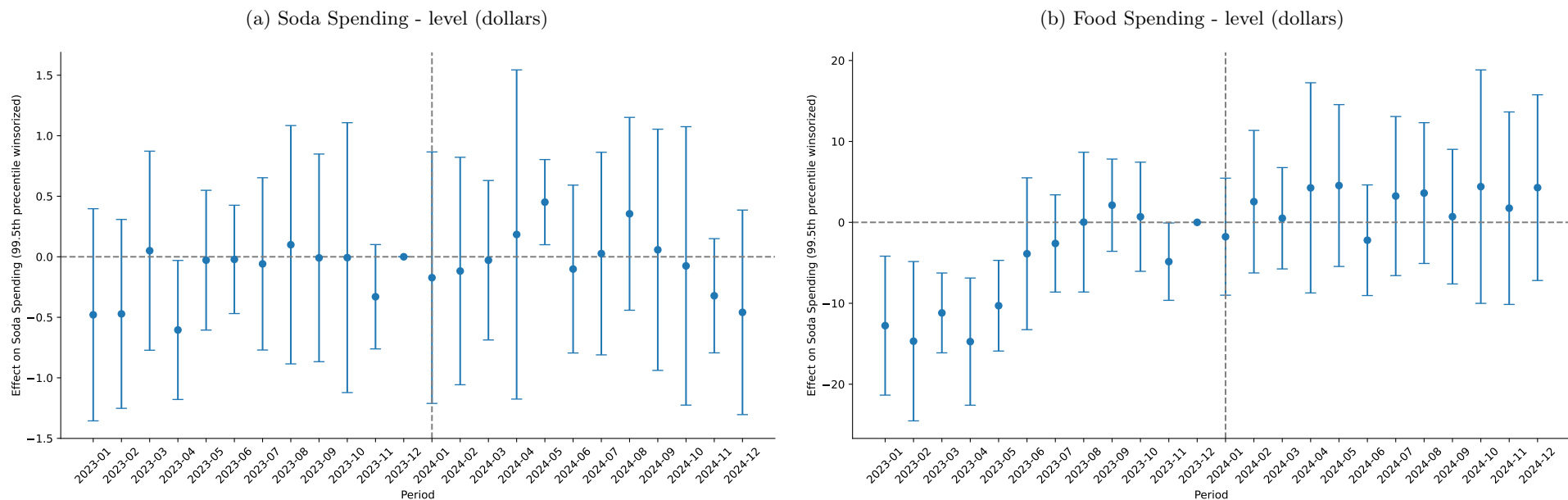


Figure 12: Event Study - TWFE (SNAP Status as Treatment within Treated States)

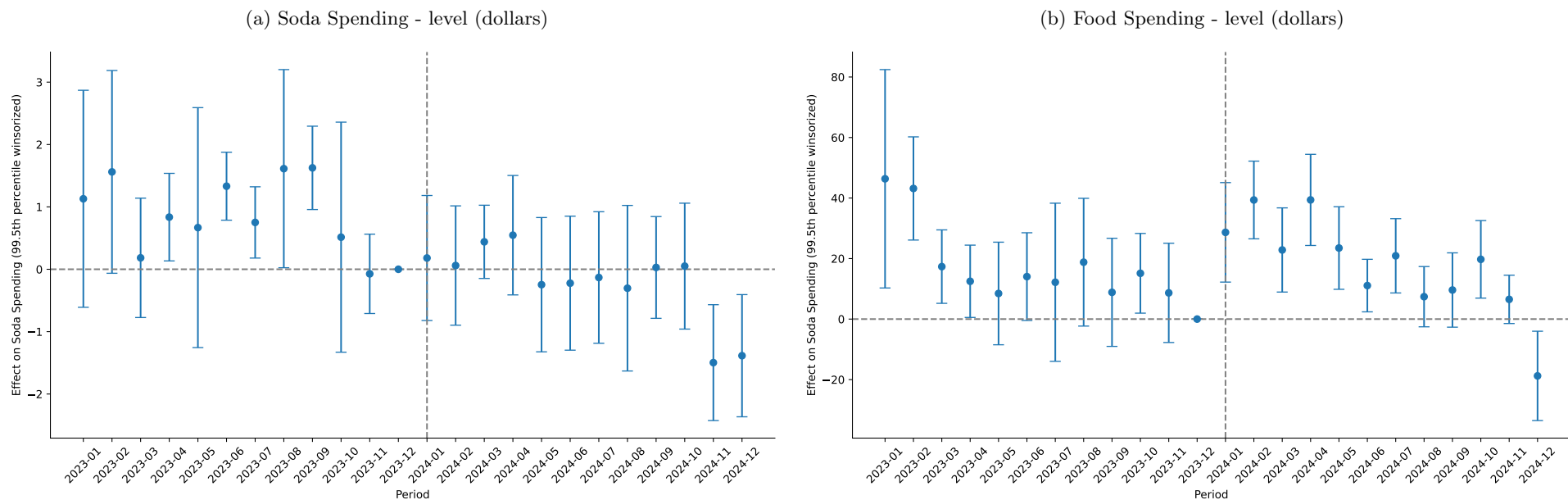
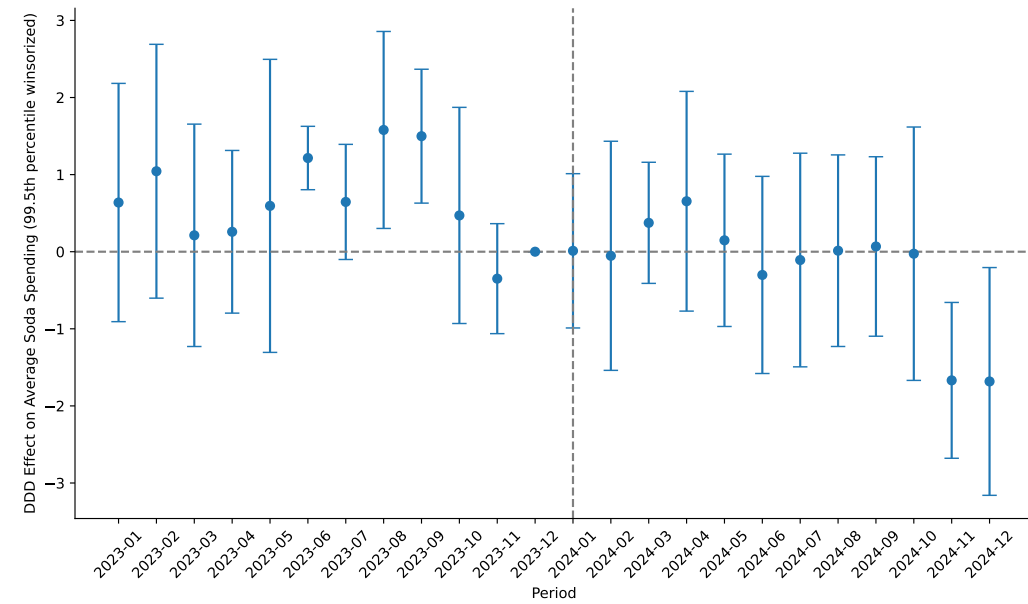
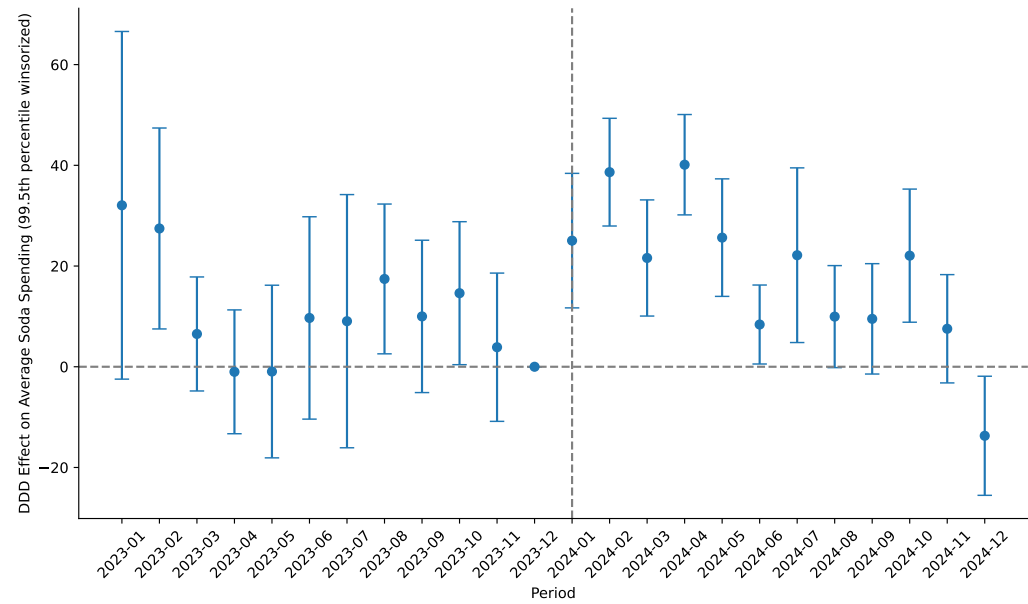


Figure 13: Event Study - DDD

(a) Soda Spending - level (dollars)



(b) Food Spending - level (dollars)



6 Regression Analysis

Table 12: Model Estimates - Soda Expenditure

Term	TWFE	TWFE (Treated State)	DDD
State $\times$ Post	0.1382 (0.1621)		
SNAP $\times$ Post		-1.0519*** (0.3198)	
State $\times$ SNAP $\times$ Post			-0.8648** (0.4012)
N	4,790,076	796,344	4,790,076
User FE	Yes	Yes	Yes
Period FE	Yes	Yes	Yes
SE Cluster	state	state	state

Notes: TWFE (SNAP) shows the estimate of TWFE with SNAP status as treatment within treated states only.

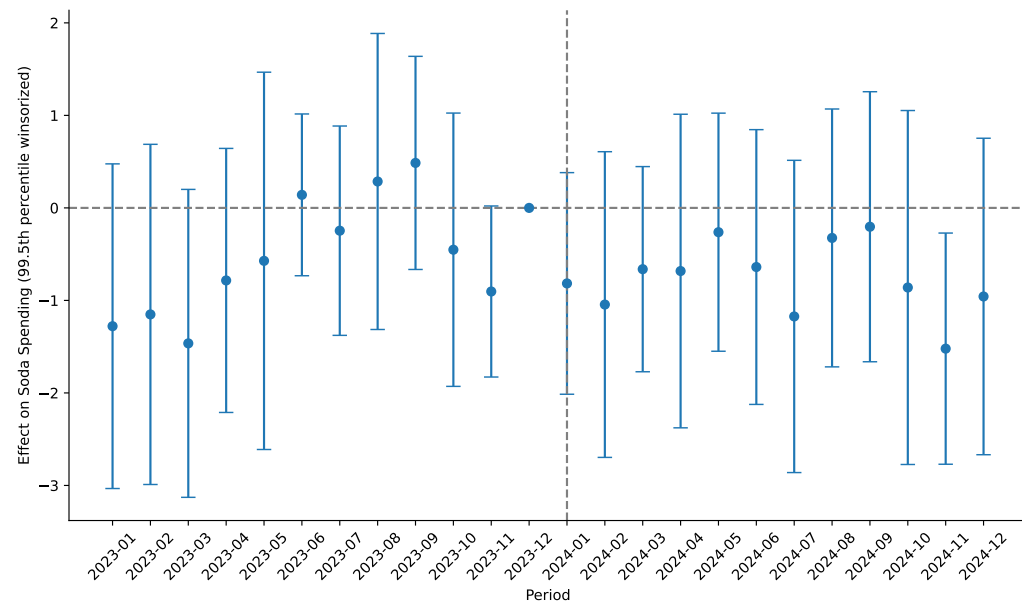
Table 13: Model Estimates - Food Expenditure

Term	TWFE	TWFE (Treated State)	DDD
State $\times$ Post	8.1754* (4.6141)		
SNAP $\times$ Post		0.3960 (4.0262)	
State $\times$ SNAP $\times$ Post			7.3516 (7.1139)
N	4,790,076	796,344	4,790,076
User FE	Yes	Yes	Yes
Period FE	Yes	Yes	Yes
SE Cluster	state	state	state

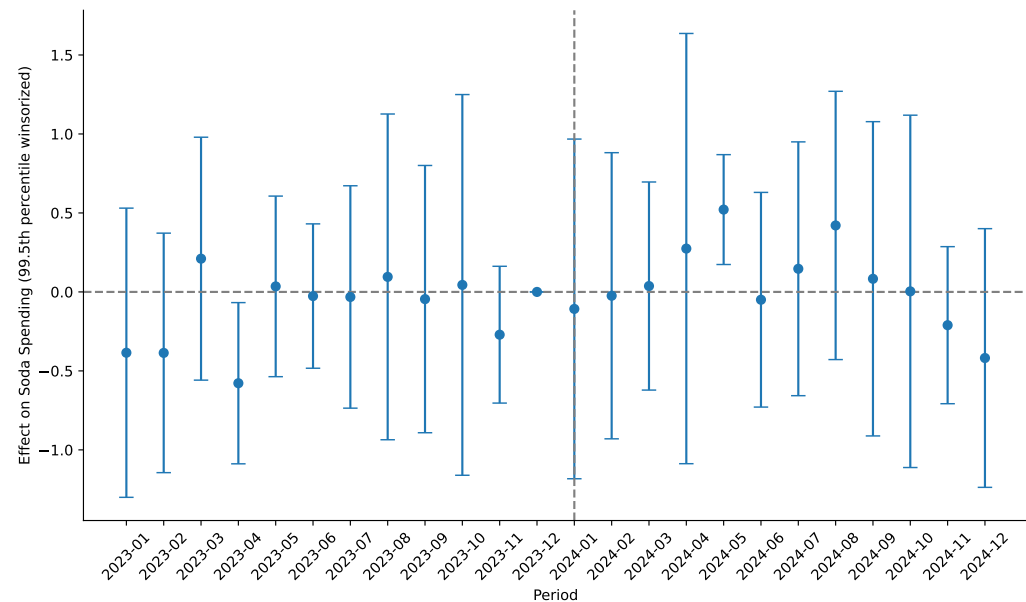
Notes: TWFE (SNAP) shows the estimate of TWFE with SNAP status as treatment within treated states only.

Figure 14: Event Study - Statement Treatment within SNAP and Non-SNAP households

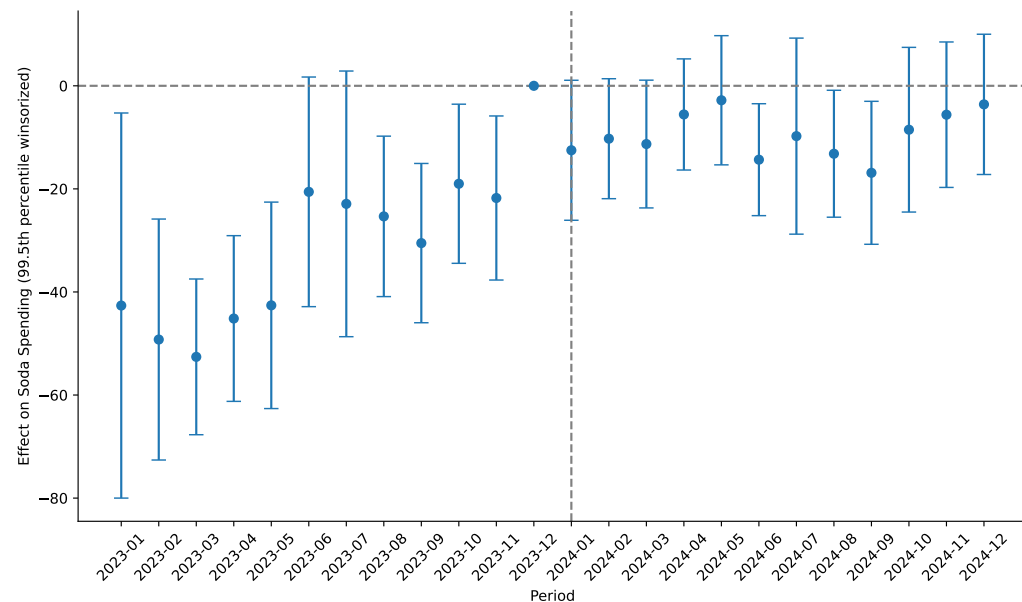
(a) Soda Spending - level (dollars) - Among SNAP households



(b) Soda Spending - level (dollars) - Among non-SNAP households



(c) Food Spending - level (dollars) - Among SNAP households



(d) Food Spending - level (dollars) - Among non-SNAP households

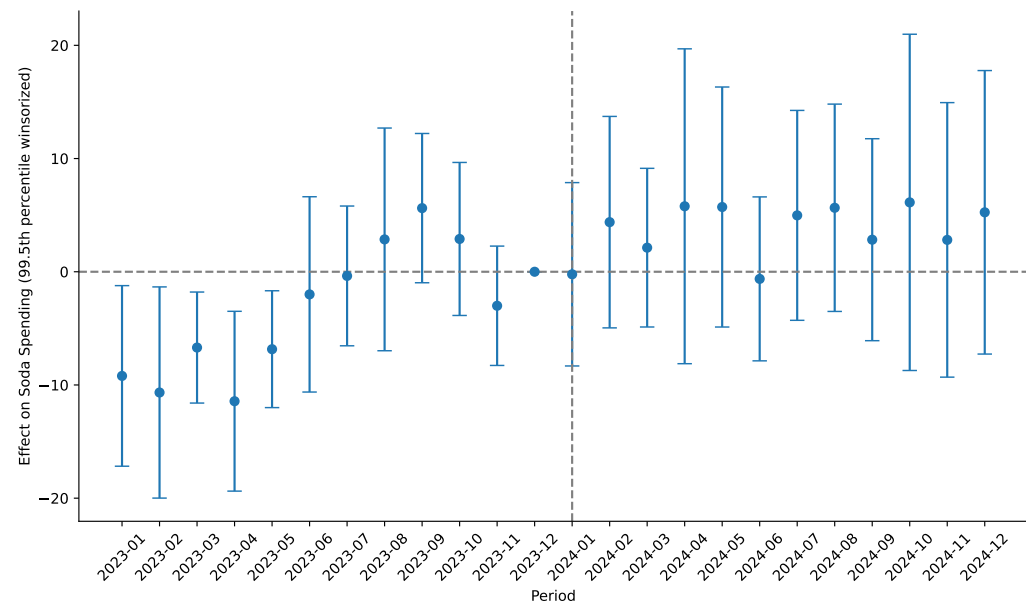


Table 14: Model Estimates - Soda Expenditure

Term	TWFE	TWFE (SNAP)	TWFE (Non-SNAP)	DDD
State $\times$ Post	0.1382 (0.1621)	-0.2674 (0.4592)	0.1679 (0.1466)	
State $\times$ SNAP $\times$ Post				-0.8648** (0.4012)
N	4,790,076	486,600	4,303,476	4,790,076
User FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes
SE Cluster	state	state	state	state

Notes: TWFE (SNAP) shows the estimate of TWFE with state as treatment within SNAP households. TWFE (Non-SNAP) shows the estimate of TWFE with state as treatment within Non-SNAP households.

Table 15: Model Estimates - Food Expenditure

Term	TWFE	TWFE (SNAP)	TWFE (Non-SNAP)	DDD
State $\times$ Post	8.1754* (4.6141)	21.5000** (8.3601)	6.9709 (4.3777)	
State $\times$ SNAP $\times$ Post				7.3516 (7.1139)
N	4,303,476	486,600	4,303,476	4,790,076
User FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes
SE Cluster	state	state	state	state

Notes: TWFE (SNAP) shows the estimate of TWFE with state as treatment within SNAP households. TWFE (Non-SNAP) shows the estimate of TWFE with state as treatment within Non-SNAP households.

Appendix

Households without Disabled or Elderly Persons (age 60 or over) Maximum Monthly Income Allowable

Household Size	1	2	3	4	5	6	7	8	9	10
Gross Monthly Income Limit (165% FPL)	\$2,152	\$2,909	\$3,665	\$4,421	\$5,177	\$5,934	\$6,690	\$7,446	\$8,203	\$8,960

For household with more than 10 persons, add \$757 for each additional person. (Updated 10/1/2025)

Age 60 & Over or Disabled Households Maximum Monthly Income Allowable

Household Size	1	2	3	4	5	6	7	8	9	10
Gross Monthly Income Limit (For those Age 60 or older OR Disabled) (200% FPL)	\$2,608	\$3,525	\$4,441	\$5,358	\$6,275	\$7,191	\$8,108	\$9,025	\$9,941	\$10,858

For each elderly and disabled household with more than 10 persons, add \$917 for each additional person. (Updated 10/1/2025)

Figure 15: SNAP Benefit - Income Threshold by Age and Household Size

Supplemental Nutrition Assistance Program Effective October 2024 Maximum Monthly SNAP Amounts

The chart below shows the maximum dollar amount of SNAP benefits your household can receive each month. This dollar amount is based on the number of people in your household, your income and your expenses. You could receive any amount up to or including these amounts.

Number of People in Your Household	Maximum Gross Monthly Benefits
1	\$ 292
2	\$ 536
3	\$ 768
4	\$ 975
5	\$ 1,158
6	\$ 1,390
7	\$ 1,536
8	\$ 1,756
9	\$ 1,976
10	\$ 2,196

For households with more than 10 persons, add \$220 for each additional person.

Figure 16: SNAP Benefit - Maximum Benefit by Household Size

Total SNAP Benefits Issued by Month vs Breakfast Cereal Buy Rate

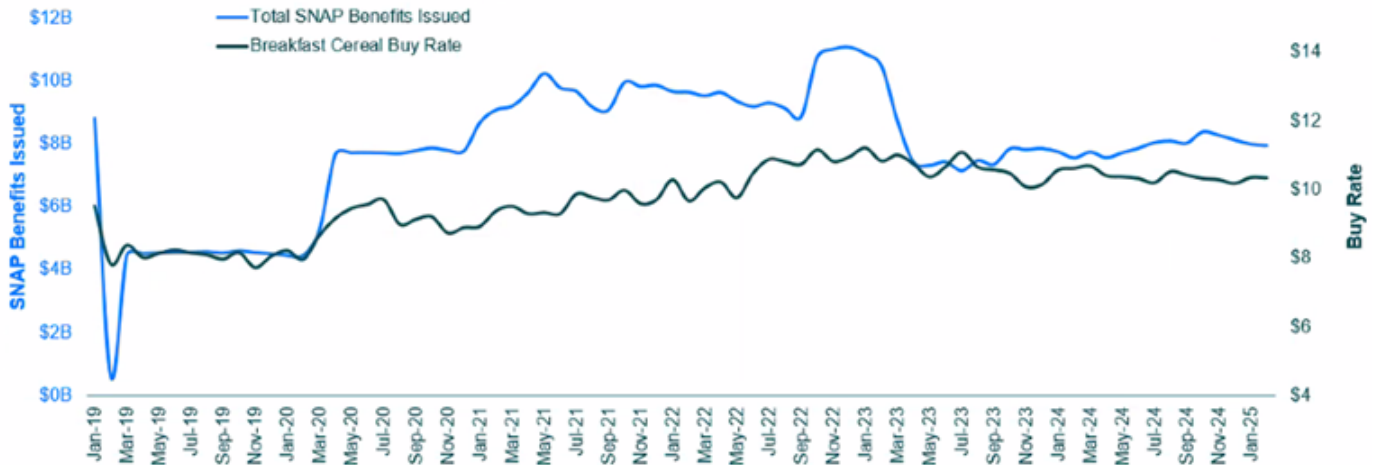


Figure 17: SNAP Benefit - Value of Total Benefit Distributed over Time

Table 16: Food at Home vs. Food Away from Home Categories

Food At Home (Grocery)	Food Away From Home (QSR)
Snack	LSR Sandwiches & Wraps
Canned	LSR Beverages
Produce	LSR Breakfast
Dairy	LSR Desserts
Beverages	LSR Entrees
Packaged Bakery (Bread & Alternatives)	LSR Snack & Sides
Condiments	LSR Mexican
Frozen Foods	LSR Sauces & Condiments
Shelf Stable Meals	LSR Italian
Candy (Snacks)	LSR Salads
Baking & Cooking	
Meat	
Seafood & Fish	
Beans & Grains	
Refrigerated Foods	
Breakfast	
Pasta & Noodles	
Deli & Prepared Foods	
In-Store Bakery (Bread & Alternatives)	
Herbs & Spices	
Alcohol Beverages	
Misc. Food	
Fresh Meal Kits	
Ice	

Table 17: Soda Categories

Soda At Home (Grocery Beverages)	Soda Away From Home (QSR Beverages)
Soft Drinks	LSR Bottled Soda
Tonic & Club Soda	LSR Fountain Drink
Sparkling Juice & Ciders	
Prebiotic & Probiotic Soda	

Figure 18: SNAP Participation Rate

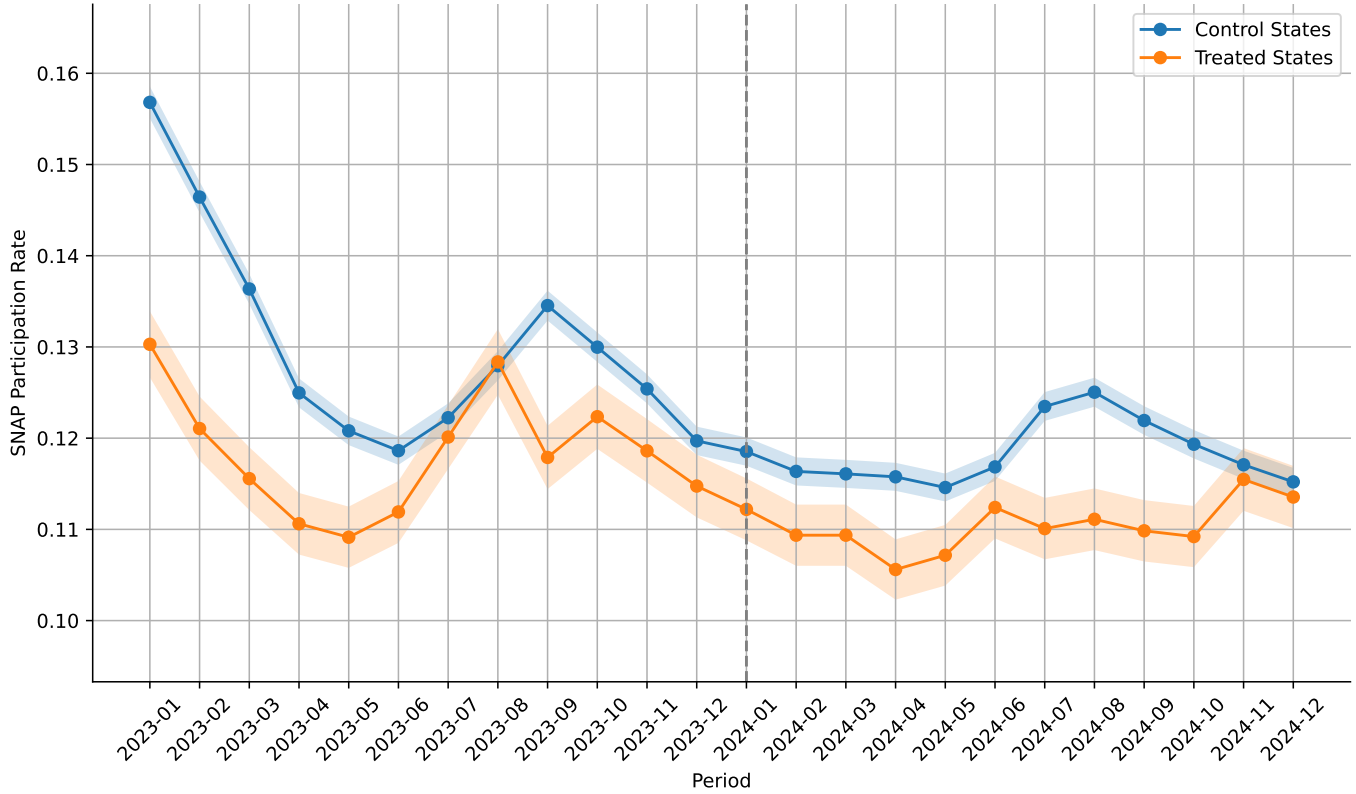


Table 18: User-Month Panel - Demographics (SNAP vs Non-SNAP)

	Mean (SD)					Min	Max
	SNAP		Non-SNAP		Total		
	Diabetes	No Diabetes	Diabetes	No Diabetes			
Income	49248.50 (48007.31)	49866.71 (49790.04)	88673.89 (58544.06)	95476.81 (64342.62)	86550.98 (62938.52)	10000.00	275000.00
Household Size	2.96 (1.65)	2.96 (1.68)	2.72 (1.36)	2.78 (1.42)	2.79 (1.45)	1.00	7.00
Age	52.80 (12.33)	48.28 (13.13)	54.12 (12.25)	49.37 (12.94)	50.59 (12.94)	19.00	67.50
Married	0.35 (0.48)	0.31 (0.46)	0.64 (0.48)	0.60 (0.49)	0.57 (0.50)	0.00	1.00
Education (years)	13.40 (1.40)	13.35 (1.44)	13.90 (1.30)	13.96 (1.35)	13.85 (1.37)	10.00	16.00
White	0.49 (0.50)	0.52 (0.50)	0.70 (0.46)	0.73 (0.44)	0.69 (0.46)	0.00	1.00
Black	0.22 (0.41)	0.20 (0.40)	0.10 (0.30)	0.08 (0.28)	0.11 (0.31)	0.00	1.00
Asian	0.05 (0.21)	0.05 (0.23)	0.06 (0.24)	0.07 (0.25)	0.06 (0.24)	0.00	1.00
Hispanic	0.22 (0.41)	0.20 (0.40)	0.12 (0.32)	0.10 (0.30)	0.12 (0.33)	0.00	1.00
Ever Child	0.28 (0.45)	0.31 (0.46)	0.20 (0.40)	0.25 (0.43)	0.25 (0.43)	0.00	1.00
Number of Users	11,037	48,414	31,465	175,490	266,406		
Average Number of Months per User	18.32	19.00	17.05	17.86	17.99		

Notes: Income's 'less than 20000' category is coded as 10000, and 'more than 250000' is coded as 275000. Other income buckets are coded as the mean of a bucket. Education's 'advanced' category is coded as 16, and 'less than high school' is coded as 10.

Table 19: User-Month Panel - Spending (SNAP vs Non-SNAP)

	Mean (SD)					Min	Max
	SNAP		Non-SNAP		Total		
	Diabetes	No Diabetes	Diabetes	No Diabetes			
Food Spending	718.31 (514.87)	675.50 (507.37)	619.19 (424.46)	585.52 (426.99)	610.31 (442.36)	0.00	50429.40
Total Spending	1595.87 (1161.51)	1507.97 (1153.81)	1565.93 (1110.22)	1485.95 (1128.68)	1513.52 (1129.46)	0.00	135684.55
Prop. of Total Spending on Food	0.48 (0.17)	0.48 (0.18)	0.43 (0.17)	0.43 (0.18)	0.44 (0.18)	0.00	1.00
Prop. of Food Spending on FAH	0.89 (0.12)	0.90 (0.13)	0.88 (0.14)	0.90 (0.13)	0.90 (0.13)	0.00	1.00
Prop. of Food Spending on FAFH	0.11 (0.12)	0.10 (0.13)	0.12 (0.14)	0.10 (0.13)	0.10 (0.13)	0.00	1.00
Prop of Food on EBT	0.33 (0.25)	0.36 (0.27)	0.16 (0.18)	0.18 (0.20)	0.30 (0.26)	0.00	1.00
Prop. of EBT Spending on FAH	1.00 (0.03)	1.00 (0.03)	1.00 (0.04)	1.00 (0.04)	1.00 (0.03)	0.00	1.00
Prop. of EBT on FAFH	0.00 (0.03)	0.00 (0.03)	0.00 (0.04)	0.00 (0.04)	0.00 (0.03)	0.00	1.00
Ever SNAP	0.82 (0.39)	0.81 (0.39)	0.07 (0.25)	0.05 (0.23)	0.18 (0.38)	0.00	1.00
Number of Users	11,037	48,414	31,465	175,490	266,406		

Notes: Prop. of food spending is calculated out of total spending. Prop. of FAH/FAFH spending and food on SNAP is calculated out of total food spending. Prop. of FAH/FAFH on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 20: User-Month Panel - Soda Spending (SNAP vs Non-SNAP)

	Mean (SD)					Min	Max
	SNAP		Non-SNAP		Total		
	Diabetes	No Diabetes	Diabetes	No Diabetes			
Soda Spending	34.25 (43.39)	29.43 (39.30)	26.61 (39.34)	20.53 (32.63)	23.67 (35.96)	0.00	5749.22
Prop. of Food Spending on Soda	0.05 (0.06)	0.05 (0.06)	0.04 (0.06)	0.04 (0.05)	0.04 (0.06)	0.00	1.00
Prop. of Soda Spending on Soda at Home	0.89 (0.23)	0.90 (0.23)	0.86 (0.27)	0.87 (0.27)	0.87 (0.27)	0.00	1.00
Prop. of Soda Spending on Soda Away From Home	0.11 (0.23)	0.10 (0.23)	0.14 (0.27)	0.13 (0.27)	0.13 (0.27)	0.00	1.00
Prop of EBT Spending on Soda	0.05 (0.11)	0.05 (0.10)	0.06 (0.15)	0.06 (0.15)	0.05 (0.12)	0.00	1.00
Prop of Soda Spending on EBT	0.30 (0.34)	0.33 (0.36)	0.15 (0.27)	0.16 (0.29)	0.28 (0.35)	0.00	1.00
Number of Users	11,037	48,414	31,465	175,490	266,406		

Notes: Prop. of soda spending is calculated out of total food spending. Prop. of soda FAH/FAFH spending is calculated out of total food spending. Prop. of soda on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 21: Annual Expenditure in 2023

	Numerator				CEX			
	100k to 149k	150k to 199k	>200k	Total	100k to 149k	150k to 199k	>200k	Total
Total Spending	18,734.63 (26,562.45)	24,901.28 (27,652.42)	25,960.63 (29,257.26)	21,546.52 (27,565.36)	90,877	117,818	174,993	127,896
Food Spending	7,255.13 (9,333.09)	9,533.27 (10,102.21)	9,906.80 (10,942.92)	8,290.33 (9,922.65)	11,980	15,264	18,525	15,256
Food at home	6,637.01 (8,736.01)	8,756.87 (9,443.60)	9,132.91 (10,189.02)	7,606.03 (9,273.25)	7,053	8,622	9,702	8,459
Food away from home	618.12 (1,153.94)	776.41 (1,318.74)	773.89 (1,618.36)	684.30 (1,298.78)	4,907	6,843	8,824	6,858
Soda spending/nonalcoholic beverage	242.78 (519.88)	295.68 (491.08)	287.38 (545.79)	263.39 (519.65)	770	856	976	867
Prop. of total spending on food	41% (0.13)	41% (0.13)	42% (0.14)	41% (0.13)	13.2%	13.0%	10.6%	12.2%
Prop. of food on FAH	91% (0.09)	92% (0.08)	92% (0.08)	92% (0.09)	58.9%	56.5%	52.4%	55.9%
Prop. of food on FAFH	9% (0.09)	8% (0.08)	8% (0.08)	8% (0.09)	41.0%	44.8%	47.6%	44.5%
Prop. of Soda/nonalcoholic beverage	3% (0.03)	3% (0.03)	3% (0.03)	3% (0.03)	6.4%	5.6%	5.3%	5.8%

Notes: This table shows the average annual spending and proportions in 2023. The soda spending row records soda spending in Numerator data, and nonalcoholic beverage spending in CEX data. Note that the proportion and total column for CEX data are computed as the proportion and average of the corresponding aggregated spending variables above.

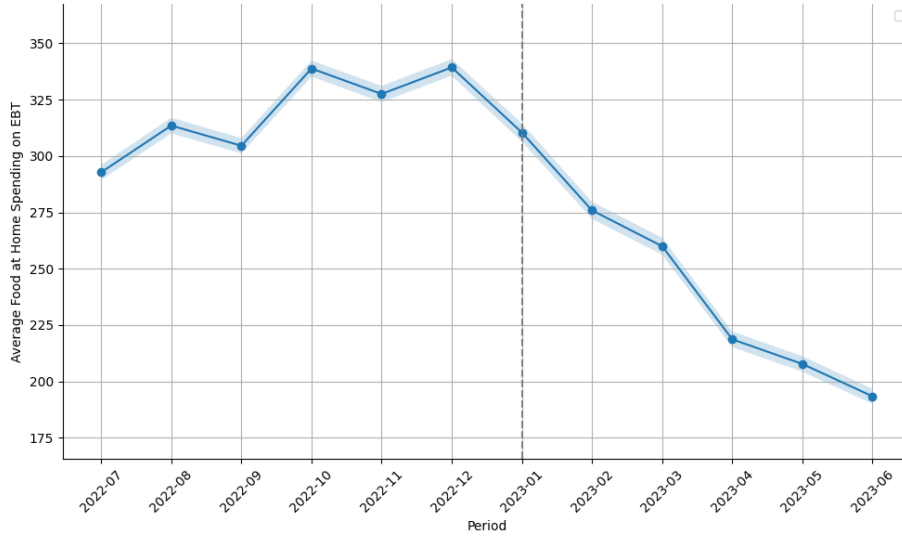


Figure 19: Household Monthly Average of Food at Home spending by EBT Card

6.1 MPC out of SNAP benefit

We estimated the MPC out of SNAP benefit during the reduction of SNAP benefit starting on Jan 2023. Figure 19 plots the trend of monthly average of food at home expenditure paid by EBT card, which starts decreasing from Jan 2023. This trend is consistent with the trend of total SNAP benefit issued based on the administrative presented in Figure 17. We first used an interrupted time series to estimate the impact of this shock on Jan 2023 on food/soda spending and food/soda EBT spending. Table 23 lays out the estimates of ITS coefficients, where period is the running variable and post is the dummy for post-Jan 2023. The reduction in SNAP benefits measured by the reduction in food EBT spending is about 42 dollars. If we take the ratio of estimates of reduction in food (-41) and food at home spending (-44) to the estimate of reduction in food EBT as a proxy for MPC out of SNAP benefit, the MPC is very close to 1, which arguable reflect the mechanical effect of a unit reduction in SNAP benefit on food/food at home consumption. This result to some extent is consistent with Song’s analysis about the effect of missing receipts that potentially reduces the crowd-out effect of a unit reduction in SNAP benefit on non-SNAP food/food at home consumption.

Table 23: Interrupted Time Series Analysis of 2023 Reduction of SNAP Benefit

	Food	Total	Food EBT	Soda	Soda EBT	FAH	FHAH	FAH EBT	Soda at Home
Intercept	724.55*** (2.44)	1503.46*** (5.36)	339.17*** (1.29)	31.33*** (0.19)	15.28*** (0.11)	669.30*** (2.31)	54.57*** (0.39)	338.63*** (1.29)	29.80*** (0.19)
period	2.78*** (0.59)	30.09*** (1.32)	-5.52*** (0.31)	0.40*** (0.05)	-0.14*** (0.03)	2.23*** (0.56)	0.54*** (0.09)	-5.50*** (0.31)	0.39*** (0.05)
post	-40.99*** (4.11)	-255.39*** (9.22)	-41.95*** (2.24)	-3.35*** (0.33)	-2.60*** (0.19)	-44.82*** (3.90)	3.73*** (0.64)	-42.02*** (2.24)	-3.47*** (0.32)
N	225244	225244	225244	225244	225244	225244	225244	225244	225244
R ²	0.00	0.00	0.02	0.00	0.01	0.00	0.00	0.02	0.00

Notes: This table presents the estimates of coefficients in an interrupted time series specification. Period is the time running variable and post the dummy variable of post-Jan 2023, where estimates of coefficients on post reveal the pre-post averages difference.

This tendency toward mechanical effect is evident in Song’s derivation in Equation

$$\begin{aligned}
\beta_{\text{SNAP}}^{\text{FAH}} &= \frac{E_i(\Delta F A H_{i,t}^{\text{true}})}{E_i(\Delta S_{i,t}^{\text{true}})} \\
&= \frac{\overbrace{E_i(\Delta S_{i,t}^{\text{true}})}^{\text{SNAP FAH expenditure}}}{E_i(\Delta S_{i,t}^{\text{true}})} + \frac{\overbrace{E_i(\Delta(F A H_{i,t}^{\text{true}} - S_{i,t}^{\text{true}}))}^{\text{Non-SNAP FAH expenditure}}}{E_i(\Delta S_{i,t}^{\text{true}})} \\
&= \underbrace{1}_{\text{mechanical effect}} + \underbrace{\frac{E_i(\Delta(F A H_{i,t}^{\text{true}} - S_{i,t}^{\text{true}}))}{E_i(\Delta S_{i,t}^{\text{true}})}}_{\text{crowd-out effect}}
\end{aligned} \tag{1}$$

1. When households tend to report more SNAP receipts than non-SNAP receipts because of Numerator's compensation scheme, $E_i(\Delta(F A H_{i,t}^{\text{true}} - S_{i,t}^{\text{true}}))$ decreases, and the estimate of MPC converges to 1. However, as will be discussed in later section, the current estimate from ITS could be confounded by the end-of-year seasonal effect, where the reduction in food consumption could be only partially explained by reduction in SNAP benefit, while the other half of the reduction could be due to the seasonally high expenditure during December and relatively normal expenditure level during January (adding DiD with other years as control to potentially partial out the seasonal effect; work in progress).